

A. Random Mood

time limit per test: 0.25 seconds
memory limit per test: 256 megabytes

This is a training contest on matrix exponentiation. See the tutorial along with hints and video solutions [here](#).

Limak is always either happy or sad. His mood switches with probability p every second. If Limak is happy right now, what is the probability that he's happy after n seconds?

Input
An integer n ($1 \leq n \leq 10^9$) and a real value p ($0 < p < 1$) with at most 9 digits after floating point.

Output
Print one line with the answer. The absolute error allowed is 10^{-6} .

Examples

input	Copy
1 0.7	
output	Copy
0.3000000000	

input	Copy
2 0.1	
output	Copy
0.8200000000	

input	Copy
11 0.06	
output	Copy
0.6225404294	

Note
In the first sample test, there's probability 0.7 that Limak changes his mood from happy to sad. Otherwise (with probability 0.3), he stays happy.

In the second sample test, the answer is $0.1 \cdot 0.1 + 0.9 \cdot 0.9 = 0.82$ because we want Limak either to switch his mood twice or not to change the mood at all.

Matrix Exponentiation

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
257540081	Apr/21/2024 12:45	Happy New Year!
257539916	Apr/21/2024 12:43	Delivered to wrong address on flight 3 WA3
257539665	Apr/21/2024 12:41	Delivered to wrong address on flight 1 WA1